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Selected Topics in Computational Quantum Physics

量子物理计算方法选讲

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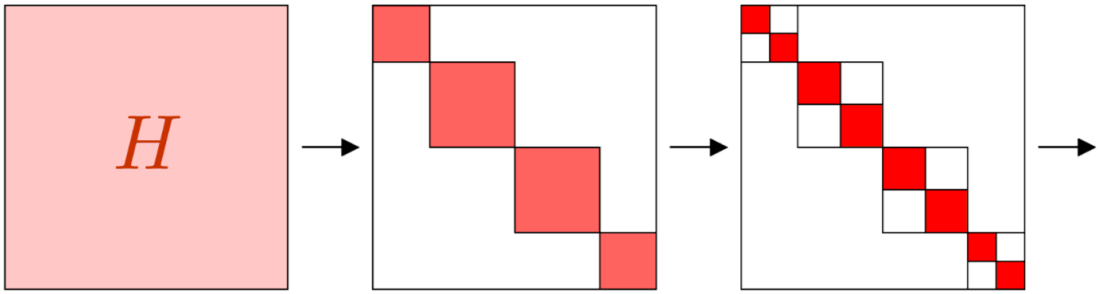
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Exact Diagonalization (ED)

$$H|\Psi\rangle = E|\Psi\rangle$$

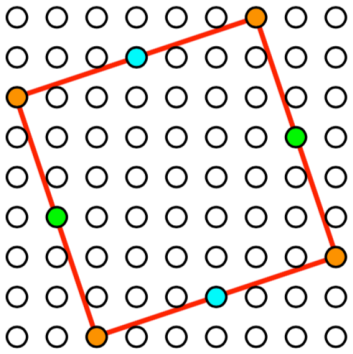
- main idea
- matrix representation of Hamiltonian
- implement symmetries
- Lanczos method
- time evolution
- finite temperature
- various applications

Symmetries

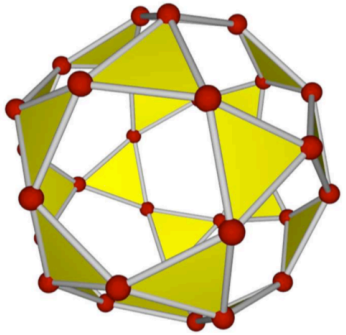


- $U(1)$ symmetry, particle number conservation
- translational symmetry, momentum space
- parity symmetry, reflection symmetry
- spin-inversion symmetry
- $SU(2)$ symmetry
- various spatial symmetry

40 sites square lattice
T ⊗ PG =40 x 4 elements



Icosidodecahedron (30 vertices)
I_h:120 elements



Dispersion relation

- total energy, momentum, and mass of particles are connected through the relativistic dispersion relation established by Paul Dirac

$$E^2 = (mc^2)^2 + (pc)^2$$

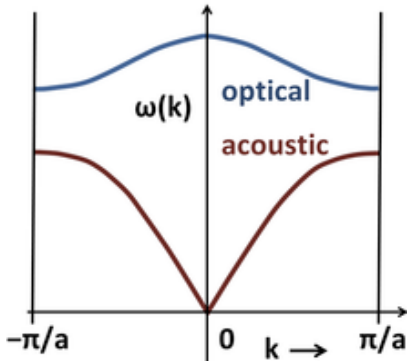
- in the ultra-relativistic limit is $E = pc$

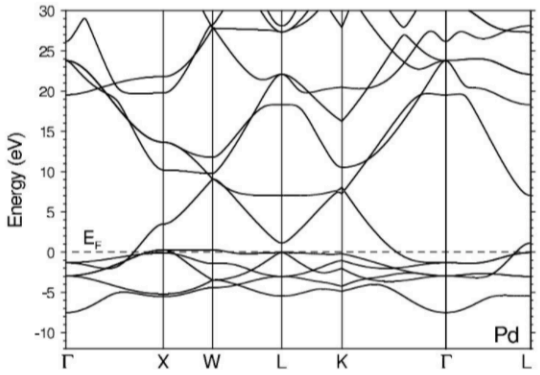
  in the non-relativistic limit is $E = p^2/2m$

- in the study of solids, the collection of all possible energies and momenta is known as the band structure of a material

  properties of the band structure define whether the material is an insulator, semiconductor or conductor

- dispersion relation determines the properties of quasi particles

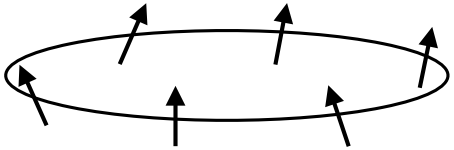




Implement translational symmetry

- using translational symmetry can further reduce the Hilbert space dimension

$$[T, H] = 0$$



T is the translational operator

$T^0|0011\rangle = |0011\rangle$, $T^1|0011\rangle = |0110\rangle$, $T^2|0011\rangle = |1100\rangle$, $T^3|0011\rangle = |1001\rangle$

- commuting operators have the same eigenstates

  idea: we first construct the eigenstates of the translational operator T,

  and then write down H on the basis of these eigenstates,

  so that H is further block-diagonalized

- the eigenstates of T can be constructed using the projector

$$P_k = \frac{1}{N}\sum_{j=0}^{N-1} e^{i\frac{2\pi}{N}kj}T^j$$

where $k = 0,\ 1,\ \cdots\ , N-1$

Eigenstate projector

- N-site ring, translating N steps brings the spins back to their original state

$$T^N = 1$$

- for a given state $|r\rangle$, the state $P_k|r\rangle$ is an eigenstate of T

$$TP_k|r\rangle = T\left[\frac{1}{N}\sum_{j=0}^{N-1} e^{i\frac{2\pi}{N}kj}T^j\right]|r\rangle = e^{-i\frac{2\pi}{N}k}\left[\frac{1}{N}\sum_{j=0}^{N-1} e^{i\frac{2\pi}{N}k(j+1)}T^{(j+1)}|r\rangle\right] = e^{-i\frac{2\pi}{N}k}P_k|r\rangle$$

the corresponding eigenvalue is $e^{-i\frac{2\pi}{N}k} = e^{-iK}$

$K = 2\pi k/N$ is the discrete lattice momentum

- P_k has the following properties

$$P_k^\dagger = \frac{1}{N}\sum_{j=0}^{N-1} e^{-i\frac{2\pi}{N}kj}T^{-j} = \frac{1}{N}\sum_{j'=0}^{N-1} e^{i\frac{2\pi}{N}kj'}T^{j'} = P_k$$

$$P_k^2 = P_k^\dagger P_k = \frac{1}{N^2}\sum_{l,j=0}^{N-1} e^{i\frac{2\pi}{N}k(l-j)}T^{(l-j)} = \frac{1}{N}\sum_{j'=0}^{N-1} e^{i\frac{2\pi}{N}kj'}T^{j'} = P_k$$

Example: $N = 4$

- for a given state $|r\rangle$, the state $P_k|r\rangle = \frac{1}{N}\sum_{j=0}^{N-1} e^{i\frac{2\pi}{N}kj}T^j|r\rangle = 0$ for some special k

	$\|r\rangle$	$T^0\|r\rangle$	$T^1\|r\rangle$	$T^2\|r\rangle$	$T^3\|r\rangle$	$P_{k=0}\|r\rangle$	$P_{k=1}\|r\rangle$	$P_{k=2}\|r\rangle$	$P_{k=3}\|r\rangle$
$\#(\uparrow) = 0$	$\|0000\rangle$	$\|0000\rangle$	$\|0000\rangle$	$\|0000\rangle$	$\|0000\rangle$	✓	×	×	×
$\#(\uparrow) = 1$	$\|0001\rangle$	$\|0001\rangle$	$\|0010\rangle$	$\|0100\rangle$	$\|1000\rangle$	✓	✓	✓	✓
$\#(\uparrow) = 2$	$\|0011\rangle$	$\|0011\rangle$	$\|0110\rangle$	$\|1100\rangle$	$\|1001\rangle$	✓	✓	✓	✓
	$\|0101\rangle$	$\|0101\rangle$	$\|1010\rangle$	$\|0101\rangle$	$\|1010\rangle$	✓	×	✓	×
$\#(\uparrow) = 3$	$\|0111\rangle$	$\|0111\rangle$	$\|1110\rangle$	$\|1101\rangle$	$\|1011\rangle$	✓	✓	✓	✓
$\#(\uparrow) = 4$	$\|1111\rangle$	$\|1111\rangle$	$\|1111\rangle$	$\|1111\rangle$	$\|1111\rangle$	✓	×	×	×

$$P_{k=1}|0101\rangle = \frac{1}{4}\left(e^{i\frac{2\pi}{4}\times 1\times 0}|0101\rangle + e^{i\frac{2\pi}{4}\times 1\times 1}|1010\rangle + e^{i\frac{2\pi}{4}\times 1\times 2}|0101\rangle + e^{i\frac{2\pi}{4}\times 1\times 3}|1010\rangle\right) = 0$$

- after translating R steps, obtaining the original state $T^R|r\rangle = |r\rangle$

  the allowed k satisfies $kR = \text{integer} \times N$ for spin systems

  $\text{mod}(kR, N) = 0$

Get properties of every configuration

- for each configuration, we need to know

(1) whether it is a representative configuration

(2) its corresponding representative configuration $|r\rangle$

(3) how many translation steps can convert it to the representative configuration

Check	0	1	2
0	1	0	0
1	1	1	0
2	0	1	1
3	1	3	0
4	0	1	2
5	1	5	0
6	0	3	1
7	1	7	0
8	0	1	3
9	0	3	3
10	0	5	1
11	0	7	3
12	0	3	2
13	0	7	2
14	0	7	1
15	1	15	0

- we write a code to generate a list called Check[i, :]
- 0th column: whether the configuration is a representative configuration, 1 for yes, 0 for no

  1st column: the corresponding representative configuration

  2nd column: the translation step R

- for example, $8 = 1000$ is obtained from the representative configuration $1 = 0001$ through 3 translation steps

  so that Check[8,0] = 0, Check[8,1] = 1, Check[8,2] = 3

Normalization and Hamiltonian matrix

- the normalized basis for each fixed-momentum Hilbert space is

$$|r_k\rangle = \frac{P_k|r\rangle}{\sqrt{\langle r|P_k^\dagger P_k|r\rangle}} = \frac{P_k|r\rangle}{\sqrt{\langle r|P_k|r\rangle}}$$

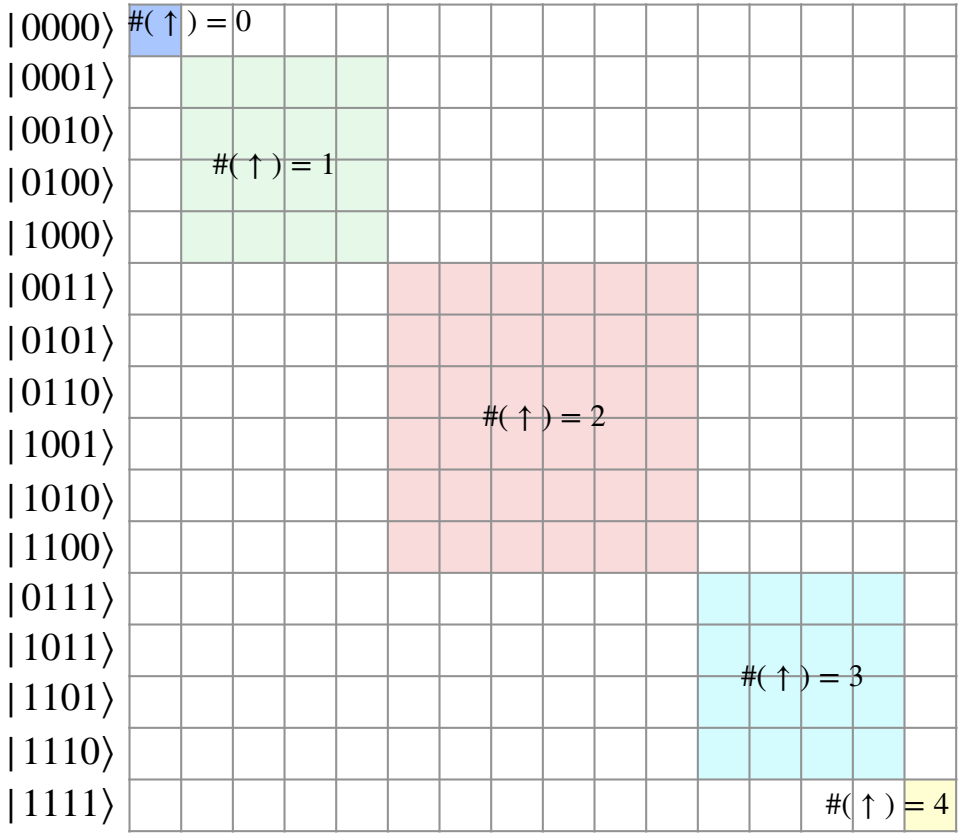
- we discard those $|r\rangle$ with $P_k|r\rangle = 0$
- since $[T, H] = 0$, we have $[P_k, H] = 0$
- we calculate the Hamiltonian matrix for a given k sector
  the matrix element between two state $|r_k\rangle$ and $|r'_k\rangle$ is given by

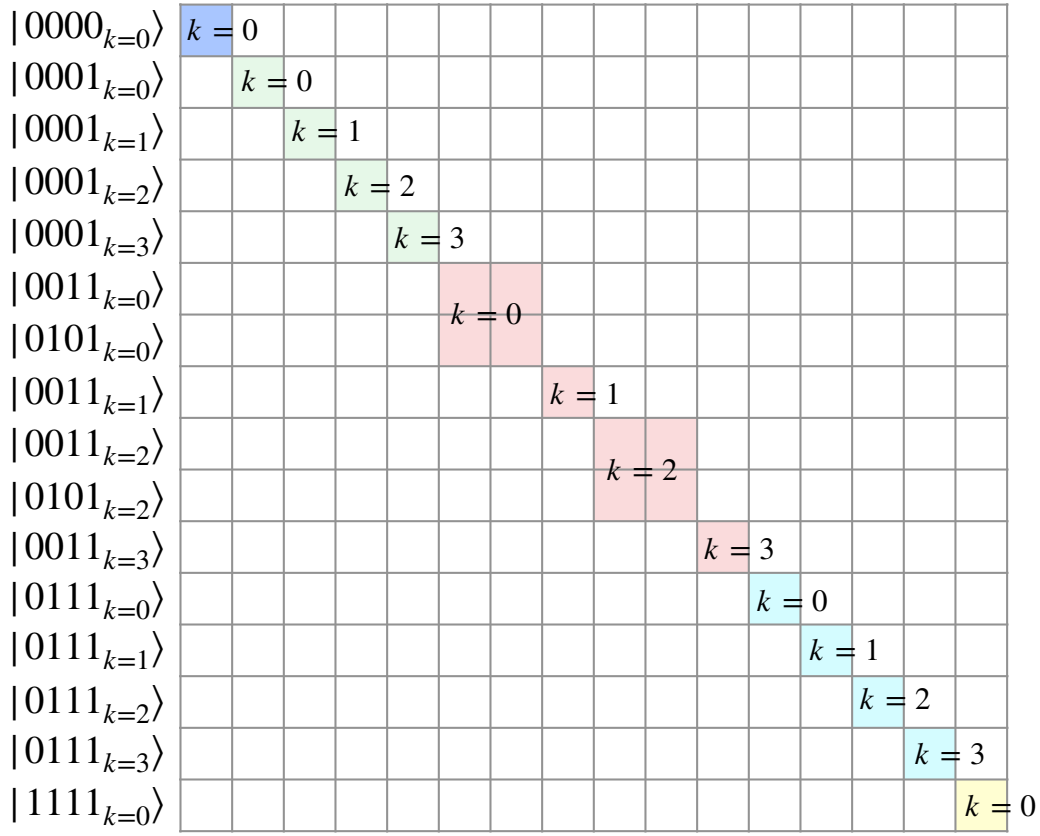
$$\langle r'_k|H|r_k\rangle = \frac{\langle r'|P_k^\dagger H P_k|r\rangle}{\sqrt{\langle r'|P_k|r'\rangle\langle r|P_k|r\rangle}} = \boxed{\frac{\langle r'|P_k H|r\rangle}{\sqrt{\langle r'|P_k|r'\rangle\langle r|P_k|r\rangle}}}$$

- we need to apply the projector only once after we applied H to the representative $|r\rangle$
  repeating the procedure for all representative $|r\rangle$, we obtain the matrix for a given k

Example: $N = 4$

- using translational symmetry can further block-diagonalized the Hamiltonian





- only $U(1)$ symmetry: $1 + 4 + 6 + 4 + 1 = 16$

  both $U(1)$ and translational symmetry:
  $1 + (1 + 1 + 1 + 1) + (2 + 1 + 2 + 1) + (1 + 1 + 1 + 1) + 1 = 16$

  only translational symmetry:
  $(1 + 1 + 2 + 1 + 1) + (1 + 1 + 1) + (1 + 2 + 1) + (1 + 1 + 1) = 16$

Be careful about fermion sign

- T translates spin-up and spin-down configurations at the same time

  T may introduce an additional sign

	spin-↓	spin- ↑	decimal	basis		
	0011	0101	53	$C^\dagger_{0\uparrow}C^\dagger_{2\uparrow}C^\dagger_{0\downarrow}C^\dagger_{1\downarrow}\|0\rangle$		
T	0110	1010	106	$C^\dagger_{1\uparrow}C^\dagger_{3\uparrow}C^\dagger_{1\downarrow}C^\dagger_{2\downarrow}\|0\rangle$		
T	1100	0101	197	$C^\dagger_{2\uparrow}C^\dagger_{4\uparrow}C^\dagger_{2\downarrow}C^\dagger_{3\downarrow}\|0\rangle$	$= C^\dagger_{2\uparrow}C^\dagger_{0\uparrow}C^\dagger_{2\downarrow}C^\dagger_{3\downarrow}\|0\rangle$	$= -C^\dagger_{0\uparrow}C^\dagger_{2\uparrow}C^\dagger_{2\downarrow}C^\dagger_{3\downarrow}\|0\rangle$
T	1001	1010	154	$C^\dagger_{3\uparrow}C^\dagger_{5\uparrow}C^\dagger_{3\downarrow}C^\dagger_{4\downarrow}\|0\rangle$	$= C^\dagger_{3\uparrow}C^\dagger_{1\uparrow}C^\dagger_{3\downarrow}C^\dagger_{0\downarrow}\|0\rangle$	$= C^\dagger_{1\uparrow}C^\dagger_{3\uparrow}C^\dagger_{0\downarrow}C^\dagger_{3\downarrow}\|0\rangle$

- therefore, we need to reconsider the allowed momentum k for a given representative configuration

  make sure $|r_k\rangle = \dfrac{P_k|r\rangle}{\sqrt{\langle r|P_k|r\rangle}}$ does not vanish

  KR may not be a multiple of N any more

How to get good quantum number?

- given a 1D finite system with translational symmetry, $[T, H] = 0$

  the crystal momentum k is a good quantum number,

  how to get E vs k in practice?

- use the aforementioned method, construct momentum basis by hand

  Good: Hilbert space is smaller; Bad: sometimes coding is a headache

- for fermionic and bosonic systems, use Fourier transformation to transform the whole Hamiltonian into momentum space

  NOT work for spin systems, why?

- get the matrix representation of T, diagonalize T to get the momentum basis

- first diagonalize H, then calculate T (diagonalize T in case of degeneracy)

- get energy and momentum simultaneously by diagonalizing HT, giving Ee^{ik}

Another possible way

- solve the eigenvalue problem

$$(H - \alpha e^{-ikT})\,|\psi\rangle = \lambda\,|\psi\rangle$$

  for the eigenvalue $\lambda = \omega - \alpha$ with the most negative real part

- it would be more elegant to consider the operator $H - \alpha e^{-ikT} - \alpha e^{ikT^\dagger}$ such that the eigenvalue problem remains Hermitian
  in practice, adding the hermitian conjugate may not needed

- when α is sufficiently large and translation symmetry is sufficiently well captured, this eigenvalue should indeed be real and correspond to an eigenstate with momentum k and energy value ω

- we need to further verify whether $|\psi\rangle$ is an eigenvector of H
  and remove the artificial ones

Transverse field Ising model

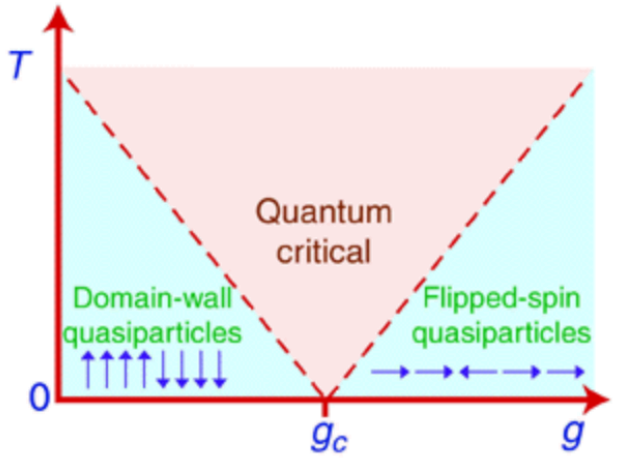
$$H = -J\sum_{i=1}^{N}\left(g\sigma_i^x + \sigma_i^z\sigma_{i+1}^z\right) \qquad J, g \geq 0$$

$$\sigma^x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \sigma^z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

- the simplest exactly solvable model with quantum phase transition
- quantum phase transition

  phase transitions at zero temperature, which occur when some parameter in the Hamiltonian is varied
- ground states in two limits

$$|G\rangle = \begin{cases} |\uparrow\uparrow\cdots\uparrow\rangle \text{ or } |\downarrow\downarrow\cdots\downarrow\rangle & g = 0 \\ |\rightarrow\rightarrow\cdots\rightarrow\rangle & g \to \infty \end{cases}$$



- $g_c = 1$ is the phase transition point
- see the book "Quantum phase transition" by Sachdev, chapter 4.2 for details

Exact solution

- Step 1: Jordan-Wigner transformation

  very powerful mapping between spin-1/2 spins and spineless fermions

we introduce $$\sigma_i^z = -\prod_{j=1}^{i-1}\left(1-2C_j^\dagger C_j\right)\left(C_i + C_i^\dagger\right), \quad \sigma_i^x = 1-2C_i^\dagger C_i.$$

then $$H = -J\sum_{i=1}^{N}\left(g\sigma_i^x + \sigma_i^z\sigma_{i+1}^z\right)$$

$$= -J\sum_{i=1}^{N} g\left(1-2C_i^\dagger C_i\right) - J\sum_{i=1}^{N-1}\left(C_i^\dagger - C_i\right)\left(C_{i+1}+C_{i+1}^\dagger\right) + JP\left(C_N^\dagger - C_N\right)\left(C_1 + C_1^\dagger\right)$$

where $$P = \prod_{i=1}^{N}\sigma_i^x = \prod_{j=1}^{N}\left(1-2C_j^\dagger C_j\right) = \exp\left(i\pi M\right), \quad M = \sum_{i=1}^{N} C_i^\dagger C_i$$

- when M is odd, $P=-1$ — periodic boundary condition

$$H = -J\sum_{i=1}^{N} g\left(1-2C_i^\dagger C_i\right) - J\sum_{i=1}^{N}\left(C_i^\dagger - C_i\right)\left(C_{i+1}+C_{i+1}^\dagger\right)$$

- when M is even, $P=+1$ — anti-periodic boundary condition

$$H = -J\sum_{i=1}^{N} g\left(1-2C_i^\dagger C_i\right) - J\sum_{i=1}^{N-1}\left(C_i^\dagger - C_i\right)\left(C_{i+1}+C_{i+1}^\dagger\right) + J\left(C_N^\dagger - C_N\right)\left(C_1 + C_1^\dagger\right)$$

Exact solution

- Step 2: Fourier transformation

we introduce

$$C_k = \frac{1}{\sqrt{N}} \sum_{j=1}^{N} C_j \exp\left(-ikj\right), \quad k = 2\pi n/N,$$

if we choose n as

$$n = \begin{cases} \text{integer} & \text{when } M \text{ is odd} \\ \text{half-integer} & \text{when } M \text{ is even} \end{cases}$$

the Hamiltonian can be written in a general form

$$H = J \sum_k \left[2\left(g - \cos k\right) C_k^\dagger C_k + i \sin k \left(C_{-k}^\dagger C_k^\dagger + C_{-k} C_k \right) - g \right].$$

Exact solution

- Step 3: Bogliubov transformation

we introduce

$$C_k = u_k\gamma_k + iv_k\gamma_{-k}^\dagger, \quad C_{-k} = u_k\gamma_{-k} - iv_k\gamma_k^\dagger$$
$$C_k^\dagger = u_k\gamma_k^\dagger - iv_k\gamma_{-k}, \quad C_{-k}^\dagger = u_k\gamma_{-k}^\dagger + iv_k\gamma_k$$

where u_k, v_k are real numbers, satisfying

$$u_k^2 + v_k^2 = 1, \quad u_{-k} = u_k, \quad v_{-k} = -v_k$$

now the Hamiltonian becomes

$$H = J\sum_k \left\{ \begin{array}{c} \gamma_k^\dagger\gamma_k \left[2\left(g - \cos k\right)u_k^2 + 2\sin k u_k v_k\right] - g \\ + \left(1 - \gamma_k^\dagger\gamma_k\right)\left[2\left(g - \cos k\right)v_k^2 - 2\sin k u_k v_k\right] \\ + \left(\gamma_k^\dagger\gamma_{-k}^\dagger - \gamma_{-k}\gamma_k\right)\left[2\left(g - \cos k\right)\left(iu_k v_k\right) - i\sin k\left(u_k^2 - v_k^2\right)\right] \end{array} \right\}$$

Exact solution

- Step 4: final step

we choose

$$u_k = \cos\left(\frac{\theta_k}{2}\right), \quad v_k = \sin\left(\frac{\theta_k}{2}\right), \quad \tan\theta_k = \frac{\sin k}{g - \cos k}$$

$$2u_k v_k = \sin\theta_k = \frac{\sin k}{\sqrt{(g-\cos k)^2 + \sin^2 k}}, u_k^2 - v_k^2 = \cos\theta_k = \frac{g - \cos k}{\sqrt{(g-\cos k)^2 + \sin^2 k}}$$

we finally have

$$H = 2J\sum_k \sqrt{1+g^2-2g\cos k}\left(\gamma_k^\dagger\gamma_k - \frac{1}{2}\right)$$

$$= 2J\sum_k \sqrt{1+g^2-2g\cos k}\gamma_k^\dagger\gamma_k - J\sum_k \sqrt{1+g^2-2g\cos k} = \boxed{\sum_k \epsilon_k \gamma_k^\dagger\gamma_k - \epsilon_0}$$

ϵ_k is the spectrum of a non-interacting single particle $\gamma_k^\dagger\gamma_k$

question: how to get many-particle eigen-energies?

show that they agree with the eigenvalues from the spin model.

Build many-particle spectrum

- $g = 2, P = +1, M$ is even, n is half-integer

$$k_{-1.5} = \frac{2\pi}{4} \times (-1.5)\,, k_{-0.5} = \frac{2\pi}{4} \times (-0.5)\,, k_{0.5} = \frac{2\pi}{4} \times (0.5)\,, k_{1.5} = \frac{2\pi}{4} \times (1.5)$$

$$\epsilon_{-1.5} = 5.59587, \epsilon_{-0.5} = 2.94725, \epsilon_{0.5} = 2.94725, \epsilon_{1.5} = 5.59587$$

$$\epsilon_0 = \frac{1}{2} \times (5.59587 + 2.94725 + 2.94725 + 5.59587) = 8.54312$$

Numerical results in $P = +1$ subspace	momentum of occupied quasi-particles	constructed many-particle eigen-energies
-8.54312	all levels empty	-8.54312
-2.64861	$(-0.5), (0.5)$	$2.94725 + 2.94725 - 8.54312 = -2.64861$
0	$(-1.5), (0.5)$	$5.59587 + 2.94725 - 8.54312 = 0$
0	$(-1.5), (-0.5)$	$5.59587 + 2.94725 - 8.54312 = 0$
0	$(1.5), (-0.5)$	$5.59587 + 2.94725 - 8.54312 = 0$
0	$(1.5), (0.5)$	$5.59587 + 2.94725 - 8.54312 = 0$
2.64861	$(-1.5), (1.5)$	$5.59587 + 5.59587 - 8.54312 = 2.64861$
8.54312	$(-1.5), (-0.5), (0.5), (1.5)$	$5.59587 + 2.94725 + 2.94725$ $+5.59587 - 8.54312 = 8.54312$

Build many-particle spectrum

- $g = 2, P = -1, M$ is odd, n is integer

$$k_{-2} = \frac{2\pi}{4} \times (-2)\,, k_{-1} = \frac{2\pi}{4} \times (-1)\,, k_0 = \frac{2\pi}{4} \times (0)\,, k_1 = \frac{2\pi}{4} \times (1)\,, k_2 = \frac{2\pi}{4} \times (2)$$

$$\epsilon_{-2} = 6, \epsilon_{-1} = 4.47214, \epsilon_0 = 2, \epsilon_1 = 4.47214, \epsilon_2 = 6$$

since $N = 4$, we can choose 4 momenta at most, either $\{k_0, k_1, k_{-1}, k_{-2}\}$or $\{k_0, k_1, k_{-1}, k_2\}$

$$\epsilon_0 = \frac{1}{2} \times (6 + 4.47214 + 2 + 4.47214) = 8.47214$$

Numerical results in P = -1 subspace	momentum of occupied quasi-particles	constructed many-particle eigen-energies
-6.47214	$[0]$	$2 - 8.47214 = -6.47214$
-4	$[-1]$	$4.47214 - 8.47214 = -4$
-4	$[1]$	$4.47214 - 8.47214 = -4$
-2.47214	$[-2]$	$6 - 8.47214 = -2.47214$
2.47214	$[-1], [0], [1]$	$4.47214 + 2 + 4.47214 - 8.47214 = 2.47214$
4	$[-2], [-1], [0]$	$6 + 4.47214 + 2 - 8.47214 = 4$
4	$[-2], [1], [0]$	$6 + 4.47214 + 2 - 8.47214 = 4$
6.47214	$[-1], [1], [-2]$	$4.47214 + 4.47214 + 6 - 8.47214 = 6.47214$

Build many-particle spectrum

- $g = 0.5$, $P = +1$, M is even, n is half-integer

$$k_{-1.5} = \frac{2\pi}{4} \times (-1.5)\,, k_{-0.5} = \frac{2\pi}{4} \times (-0.5)\,, k_{0.5} = \frac{2\pi}{4} \times (0.5)\,, k_{1.5} = \frac{2\pi}{4} \times (1.5)$$

$$\epsilon_{-1.5} = 2.79793, \epsilon_{-0.5} = 1.47363, \epsilon_{0.5} = 1.47363, \epsilon_{1.5} = 2.79793$$

$$\epsilon_0 = \frac{1}{2} \times (2.79793 + 1.47363 + 1.47363 + 2.79793) = 4.27156$$

Numerical results in $P = +1$ subspace	momentum of occupied quasi-particles	constructed many-particle eigen-energies
-4.27156	all levels empty	-4.27156
-1.32431	$(-0.5), (0.5)$	$1.47363 + 1.47363 - 4.27156 = -1.32431$
0	$(-1.5), (0.5)$	$2.79793 + 1.47363 - 4.27156 = 0$
0	$(-1.5), (-0.5)$	$2.79793 + 1.47363 - 4.27156 = 0$
0	$(1.5), (-0.5)$	$2.79793 + 1.47363 - 4.27156 = 0$
0	$(1.5), (0.5)$	$2.79793 + 1.47363 - 4.27156 = 0$
1.32431	$(-1.5), (1.5)$	$2.79793 + 2.79793 - 4.27156 = 1.32431$
4.27156	$(-1.5), (-0.5), (0.5), (1.5)$	$2.79793 + 1.47363 + 1.47363 + 2.79793 - 4.27156 = 4.27156$

- $g = 0.5$, $P = -1$, M is odd, n is integer

$$k_{-2} = \frac{2\pi}{4} \times (-2)\,, k_{-1} = \frac{2\pi}{4} \times (-1)\,, k_0 = \frac{2\pi}{4} \times (0)\,, k_1 = \frac{2\pi}{4} \times (1)\,, k_2 = \frac{2\pi}{4} \times (2)$$

$$\epsilon_{-2} = 3, \epsilon_{-1} = 2.23607, \epsilon_0 = 1, \epsilon_1 = 2.23607, \epsilon_2 = 3$$

since $N = 4$, we can choose 4 momenta at most, either $\{k_0, k_1, k_{-1}, k_{-2}\}$or $\{k_0, k_1, k_{-1}, k_2\}$

$$\epsilon_0 = \frac{1}{2} \times (3 + 2.23607 + 1 + 2.23607) = 4.23607$$

Numerical results in P = -1 subspace	momentum of occupied quasi-particles	constructed many-particle eigen-energies
-4.23607	all levels empty	-4.23607
-1	$[-1], [0]$	$2.23607 + 1 - 4.23607 = -1$
-1	$[1], [0]$	$2.23607 + 1 - 4.23607 = -1$
-0.23607	$[-2], [0]$	$3 + 1 - 4.23607 = -0.23607$
0.23607	$[-1], [1]$	$2.23607 + 2.23607 - 4.23607 = 0.23607$
1	$[-1], [2]$	$2.23607 + 3 - 4.23607 = 1$
1	$[1], [-2]$	$2.23607 + 3 - 4.23607 = 1$
4.32607	$[-1], [0], [1], [2]$	$2.23607 + 1 + 2.23607$ $+3 - 4.23607 = 4.32607$

Homework rules

- this homework has one task
  please submit through http://learn.tsinghua.edu.cn

- please submit source code and a detailed note to explain your code and results

- deadline is Oct 20, 23:00

- if you submit after the deadline, you will have less points

- if you are an expert in exact diagonalization, this homework may be waived
  please contact me in person and submit something you have done before

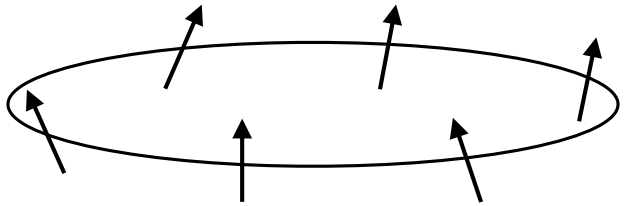
Homework

- consider the 1D spin model on a N-site ring

  please take parameters: $g = 1$, $J = 1$, $h = 1$

$$H = -\sum_j \left(g\sigma_{j-1}^z \sigma_j^y \sigma_{j+1}^z + J\sigma_j^z \sigma_{j+1}^z + h\sigma_j^x \right)$$

  where σ^x, σ^y, and σ^z are Pauli matrices



- write a code to calculate the dispersion relation $E(k)$ of this model

  show the eigenvalues in each momentum sector for $N = 8$

- calculate the ground state energy E per site and the magnetization per site $\langle \sigma_i^z \rangle$ and $\langle \sigma_i^x \rangle$ using the true ground state

Sample output

- please note the input parameters are different from homework

```
Case: g= 1 J= 1 h= 1.2 Ns= 6

(1)_Momentum_Sector_ki= 0 : [-9.6871158  -9.35916998 -6.67892917 -5.47466807 -3.13404403 -0.80328314
  0.3865042   1.20960813  2.          2.54105608  2.74203618  4.48321614
  8.58816183  9.18662763]
(1)_Momentum_Sector_ki= 1 : [-6.99254762 -5.49321515 -3.42237678 -0.50509985  0.3877056   1.24801119
  2.33479699  4.77600193  5.66672369]
(1)_Momentum_Sector_ki= 2 : [-6.65776694 -4.32572458 -2.87321219 -2.15188254 -0.77579702 -0.32347021
  0.27409269  2.72558454  3.39083412  4.91944974  7.79789239]
(1)_Momentum_Sector_ki= 3 : [-4.75608342 -4.1176038  -3.7094474  -2.         -2.          0.32957255
  2.02823755  3.7094474   5.76674783  8.74912928]
(1)_Momentum_Sector_ki= 4 : [-6.65776694 -4.32572458 -2.87321219 -2.15188254 -0.77579702 -0.32347021
  0.27409269  2.72558454  3.39083412  4.91944974  7.79789239]
(1)_Momentum_Sector_ki= 5 : [-6.99254762 -5.49321515 -3.42237678 -0.50509985  0.3877056   1.24801119
  2.33479699  4.77600193  5.66672369]
(2)_Ground_State_energy_per_site= -1.6145193003964797
(2)_Ground_State_sigmax_per_site= (0.6029072339605066+0j)
(2)_Ground_State_sigmaz_per_site= (-2.146431180941969e-15+0j)
```

Grading of this homework

- academic integrity [+50]

  completing assignments independently, creating and expressing your own ideas, DON'T copy answers from others or allow others to copy your answers

- the source code can be executed [+10] and can provide correct results [+10]

  the source code has high readability [+10]

- there is a detailed note file to explain the basic principle, source code and results [+10]

  the note file is well-written and easy to understand [+10]

- did not show the eigenvalues in each momentum sector [-20]

  did not show the magnetization [-20]

- time-dependent score: after DDL: [-1 per day]

- self-motivation score: new ideas or extra results [+10]